

Lying Latents: Auditing Distribution-Free Inference for Sparse Autoencoder Feature Discovery

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1 The Problem

Interpretability literature often asks whether a discovered feature is *real*. The standard workflow is: train a sparse autoencoder (SAE) on a language model’s activations, search its latents for one that correlates with a concept of interest, then validate that latent by ablating or steering it and observing a behavioural change.

Consider the search-validate case: You have some 16,000 latents and no labels. Which latent corresponds to some chosen concept?

Search: Run a bunch of text through the model, check which latents activate. Say we’re trying to track the concept of sarcasm. We scan all 16,000 latents to find one whose activation pattern correlates the most with the presence of sarcasm. Say latent #K correlates with 0.7.

Validate: Intervene. Clamp that latent #K low or high, check new model behaviour. If expected behaviour moves, you claim you’ve shown causality, not just correlation. You’d mark latent #K as the sarcasm feature.

This is selective inference. There are two main problems with this:

Problem #1: Multiple testing. You checked 16k latents. At a 5% significance threshold, you pass 800 latents purely by random chance.

Problem #2: Overlapping search and validate sets. You used some data to pick latent #K and then validated if it worked or not on the same data. Consider the case where latent #K’s correlation of 0.7 was 0.4 real and 0.3 noise. Re-measuring on the same data would retain that same 0.7 correlation, because the noise still persists.

Does causality not save us? Ablating a latent that was chosen by chance still perturbs the model. Something will move. You don’t know what a null interven-

tion looks like (when you intervene on a genuinely null latent).

The field knows and acknowledges this, but it has only recently started importing statistical machinery to fix it: false discovery rate (FDR) control, knock-offs, conformal methods. A recent instance applies Model-X knockoffs (Candès et al., 2018) to SAE latents and reports FDR-controlled selection, concluding that roughly a quarter of high-activity latents carry task-relevant signal and the rest are noise (Enkhbayar, 2025). Its finite-sample guarantee is instantiated *via a Gaussian surrogate for the latent distribution*. More specifically: the knockoffs procedure requires generating a sort of “decoy” for each latent, by sampling from that latent’s distribution conditional on all the other latents. Current work assumes this conditional distribution is Gaussian.

This is where the project starts. How are SAE latents distributed? They’re non-negative under ReLU/JumpReLU (Rajamanoharan et al., 2024), and heavy-tailed. Sparsity means it is exactly zero the majority of the time. A Gaussian assumption isn’t entirely valid, but there is a justifiable (kind of) reason to use it anyway: FDR control is robust to getting the distribution somewhat wrong, and the inflation is bounded by how far off you are, measured in KL divergence (Barber et al., 2020). Here is why that defence fails. SAE latents are exactly zero most of the time, whereas a Gaussian assigns zero probability to an exact zero. This means the true distribution is not absolutely continuous with respect to the Gaussian model, so the KL divergence is actually infinite rather than just large. In other words, the KL-based bound cannot be applied here. But this only kills a guarantee. It does not actually tell us if the procedure is actually bad. Whether it actually does is an empirical question, and measuring it is the first thing we do.

2 Why It Matters

Interpretability’s central weakness is that it is hard to say whether an interpretability claim is correct. We hypothesize that this is evidence of a missing statistical protocol rather a bad dictionary-learning objective.

3 The Plan

1. **Measure the damage: a shuffled-label null.** Randomly permute labels. Destroy genuine feature-label relationships. Run standard pipeline on this meaningless data. Find best latent(s). Ablate. Check behaviour changes. Ideally, the pipeline should find nothing. Whatever rate at which it does find things at is its FDR; measured directly.
2. **Audit the knockoff assumption.** Test the Model-X exchangeability property on real Gemma Scope and Pythia SAE latents. Exchangeability essentially just says that the decoy must be statistically indistinguishable from the real copy. But this holds iff you sample from the correct conditional law. Assume Gaussian, you only get a second order knockoff. First two moments match with underlying SAE latents, then nothing else.
3. **Supply a procedure that holds.** Fix it. Working on this.

4 Downstream Check: Does Survival Predict Steering?

A filter that only certifies itself is a formality, so we test it against a downstream NLP task. Feature *steering* (clamping a latent high to shift generation) is how SAE features actually get used, and it is known to be unreliable, with the literature openly disagreeing about whether SAE features steer better than simple baselines. We ask whether features surviving our filter are the ones that steer.

5 Feasibility

Public SAEs (Lieberum et al., 2024) on Gemma-2-2B and Pythia; sentiment (SST-2) and templated prompts; semi-synthetic planted-signal data for FDR ground truth. No SAE or language-model training is needed: everything is forward passes plus CPU statistics, so a single mid-range GPU suffices (hopefully).

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